

AWS Hotstar Project - Full DevOps Guide (Errors + Code + Commands + Architecture)

1. Common Errors + Fixes

	Cause	Fix
in requires at least 1 argument	Broken multiline script in Launch Template	Use correct image name + safe multiline OR single line c
roup unhealthy	Wrong health check or app not ready	Ensure /health returns 200 + add sleep 10-15 sec
out	SG/NACL/Public IP missing	Allow port 22, attach public IP, correct route to IGW
ll denied	Docker Hub auth or wrong repo	docker login + correct image name

2. Architecture Flow

User → ALB → Target Group → EC2 (ASG) → Docker App (/health)

Private subnet uses NAT Gateway for internet access

IGW handles inbound/outbound routing for public subnet

3. Flask App Code (app.py)

```
from flask import Flask, jsonify
import os, requests

app = Flask(__name__)

API_KEY = os.getenv("TMDB_API_KEY")

BASE_URL = "https://api.themoviedb.org/3"

@app.route('/')
def home():
    return "Hotstar App Running"

@app.route('/health')
def health():
    return jsonify(status="ok"), 200

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=80)
```

4. Docker Commands

```
# Build image
docker build -t hotstar-app .

# Tag image
docker tag hotstar-app username/hotstar-app

# Login Docker Hub
docker login

# Push image
docker push username/hotstar-app

# Run container
docker run -d -p 80:80 --name hotstar-app --restart always -e TMDB_API_KEY=your_key username/hotstar-app

# Check running containers
docker ps
```

5. AWS / Linux Commands

```
# EC2 connect
ssh -i key.pem ubuntu@<public-ip>

# Check logs
cat /var/log/cloud-init-output.log

# Check Docker logs
docker logs hotstar-app

# Check health
curl localhost/health

# Restart Docker
systemctl restart docker
```

6. Launch Template User Data Script

```
#!/bin/bash
apt update -y
apt install -y docker.io

systemctl start docker
systemctl enable docker

sleep 15

docker pull username/hotstar-app

docker run -d -p 80:80 --name hotstar-app --restart always -e TMDB_API_KEY=your_key username/hotstar
```

7. Key Learning Summary

- Always verify Launch Template version in ASG
- Always ensure /health endpoint returns 200
- Docker image must exist in registry before run
- Add sleep in user-data for app readiness
- Networking (SG + RT + IGW + NAT) must align properly